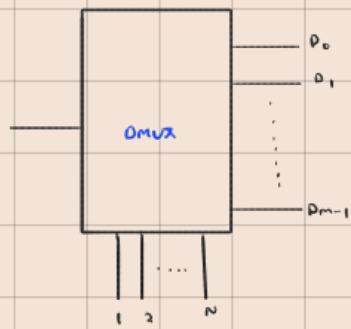
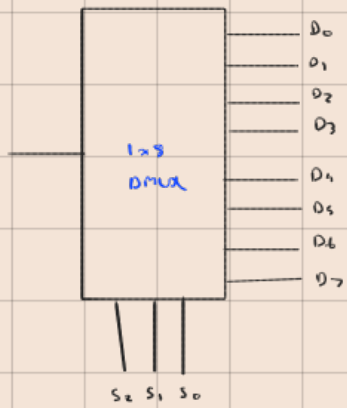
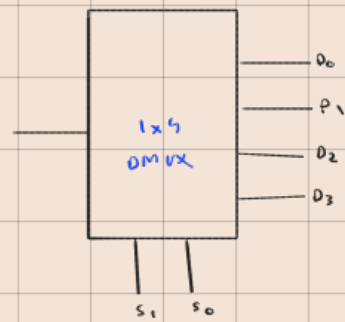
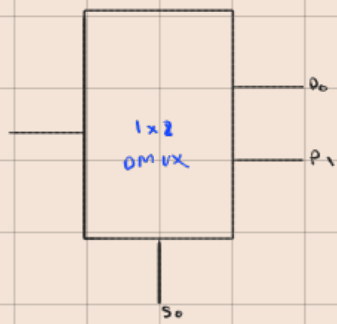


De multiplexer

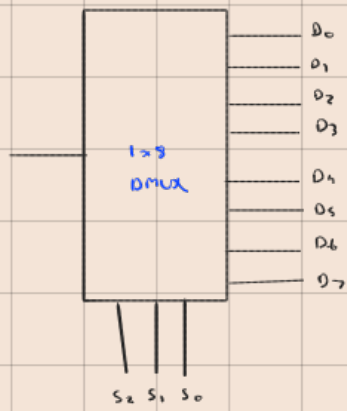


$$M = 2^N$$

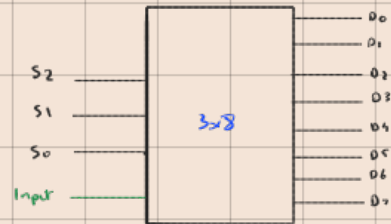
$$N = \log_2(M)$$



1 to 8 DMUX

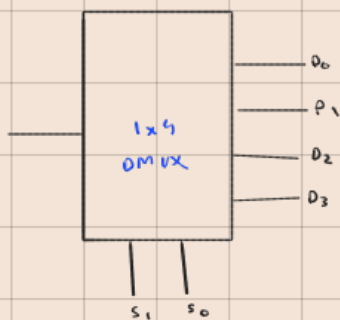


- Same as 3x8 Decoder



S_2	S_1	S_0	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1							
0	0	1		1						
0	1	0			1					
0	1	1				1				
1	0	0					1			
1	0	1						1		
1	1	0							1	
1	1	1								1

1 to 4 Demultiplexer

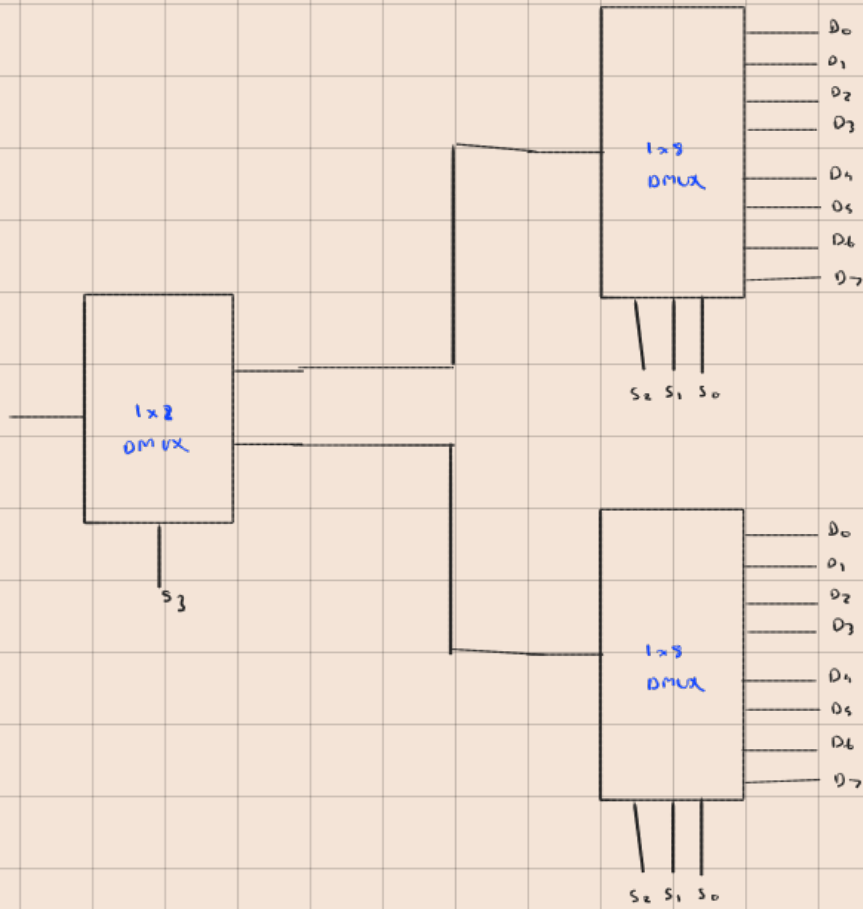


2x4 Decoder

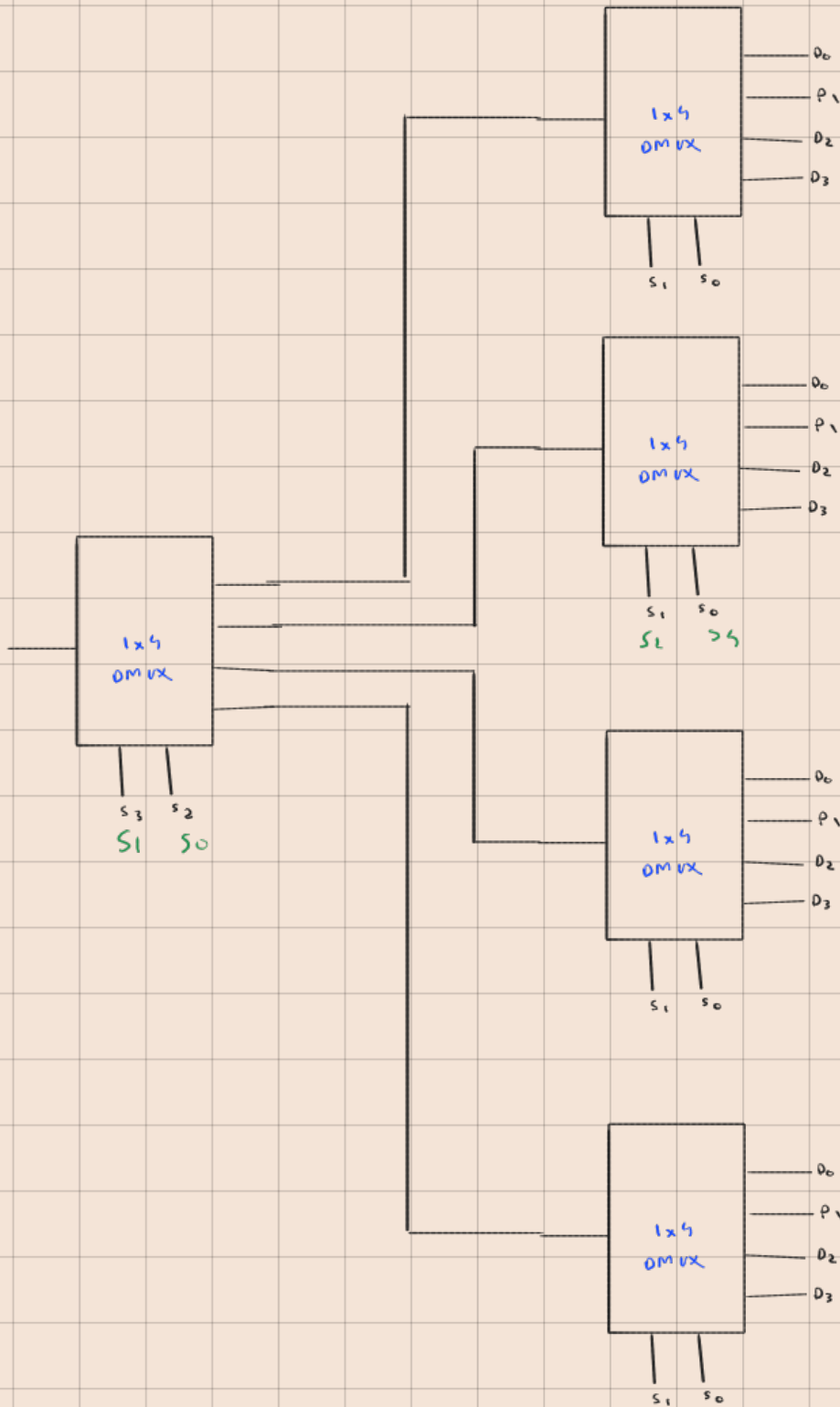
S_1	S_0	D_0	D_1	D_2	D_3
0	0	1			
0	1		1		
1	0			1	
1	1				1

- we can use 2x4 Decoder

1 x 16 using 1 x 8



1x16 using 1x4

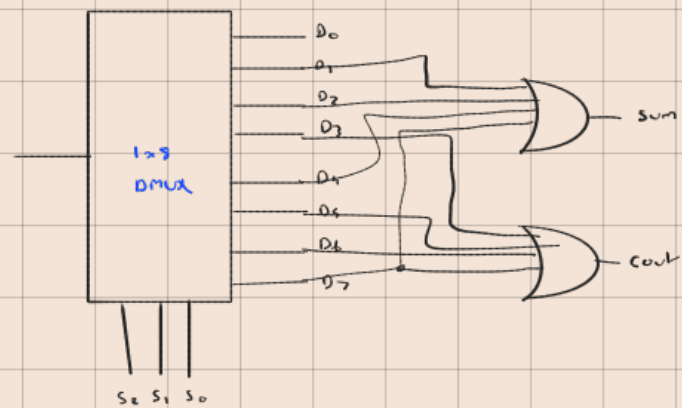


Logic Circuit Implementation

- To implement a logic circuit we use n variables. we use n input pins demultiplexer.
- If 3 variables then, 3 selection pins or 1×8 multiplexer

Full Adder-

A	B	C _{in}	Sum	Carry	
0	0	0	0	0	P ₀
0	0	1	1	0	P ₁
0	1	0	1	0	P ₂
0	1	1	0	1	P ₃
1	0	0	1	0	P ₄
1	0	1	0	1	P ₅
1	1	0	0	1	P ₆
1	1	1	1	1	P ₇



This is the same as 3×8 Decoder. The Input pin in a multiplexer is the Enable pin which has to be 1 in order for the three selecting pins to generate an output.